58 – Health Insurance System

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1. Introduction

The Health Insurance Management System is designed to digitize the process of managing medical insurance balances and healthcare payments. The system allows patients to subscribe to insurance plans through a simulated payment process. Healthcare provider registered in the system can submit treatment information for patients, and the system automatically deducts the treatment cost from the patient's insurance balance according to their plan coverage.
The system reduces manual paperwork, speeds up payment processing, and ensures accurate tracking of medical expenses. It supports healthcare organizations by providing an efficient and reliable platform for managing insurance coverage and financial transactions between patients and hospitals.

2. User Requirements

Users require a system that:

- Allow patients to register and subscribe to insurance plans.
- Allows admin to add hospitals to the system.
- Allows Healthcare provider to submit patient treatment details.
- Automatically deducts treatment costs from patient balance.
- Allow patients to view their remaining insurance balance.
- Ensures secure login for different types of users.
- Provides fast and reliable system performance.
- Prevents treatment submission if balance is insufficient.

3. Functional Requirements

3.1 Functional Requirement 1: Patient Registration

- Description/ Action: Patient creates an account in the system.
- Requirements/ Inputs: Full name, national ID, email, password.
- Source: Patient.
- Pre-condition: Patient does not already have an account.
- Post-condition: Patient account is saved in the system database.
- Output: Confirmation message indicating successful registration.

3.2 Functional Requirement 2: Insurance Plan Subscription

- Description/ Action: Patient selects an insurance plan and completes a simulated payment process to activate insurance coverage.
- Requirements/ Inputs: Selected plan, payment details.
- Source: Patient.
- Pre-condition: Patient account exists.

- Post-condition: Insurance balance is assigned based on selected plan.
- Output: Confirmation message showing activated insurance plan and available balance.

3.3 Functional Requirement 3: Admin Adds Healthcare provider

- Description/ Action: Admin registers hospitals in the system to allow them to submit treatment information.
- Requirements/ Inputs: Healthcare provider name, Healthcare provider ID, contact information.
- Source: Admin.
- Pre-condition: Admin is authenticated.
- Post-condition: Healthcare provider is stored in system database.
- Output: Confirmation message indicating Healthcare provider successfully added.

3.4 Functional Requirement 4: Healthcare provider Submits Patient Treatment Information

- Description/ Action: Healthcare provider enters patient treatment details including cost of medical service.
- Requirements/ Inputs: Patient national ID, treatment cost.
- Source: Healthcare provider.
- Pre-condition: Healthcare provider exists in system and patient has active insurance plan.
- Post-condition: Treatment record stored in database.
- Output: Confirmation message showing successful submission.

3.5 Functional Requirement 5: Automatic Deduction from Insurance Balance

- Description/ Action: System automatically deducts treatment cost from patient insurance balance after healthcare provider submits treatment.
- Requirements/ Inputs: Treatment cost, patient insurance balance.
- Source: System database.
- Pre-condition: Patient balance is greater than or equal to treatment cost.
- Post-condition: Updated balance stored in system.
- Output: Updated balance displayed.

3.6 Functional Requirement 6: View Remaining Insurance Balance

- Description/ Action: Patient views available insurance balance after deductions.
- Requirements/ Inputs: Patient ID.
- Source: System database.
- Pre-condition: Patient is logged in.
- Post-condition: Balance retrieved from system.

- Output: Display of remaining insurance balance.

4. Non-Functional Requirements

4.1 Non-Functional Requirement 1: Performance

The system should process user requests and display results within 2 seconds under normal conditions.

4.2 Non-Functional Requirement 2: Security

User passwords must be encrypted before storing them in the database. Only authorized users should access system features.

4.3 Non-Functional Requirement 3: Reliability

The system should prevent incorrect operations such as submitting treatment information when the patient balance is insufficient.

4.4 Non-Functional Requirement 4: Usability

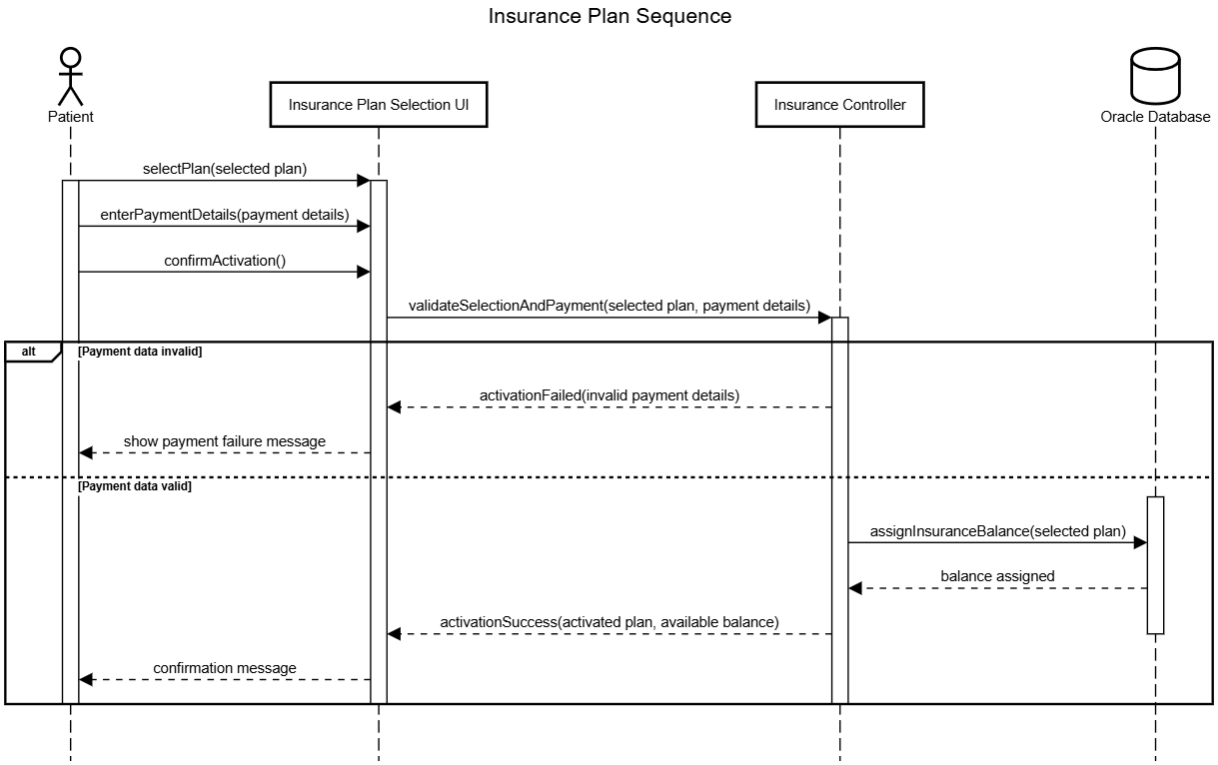
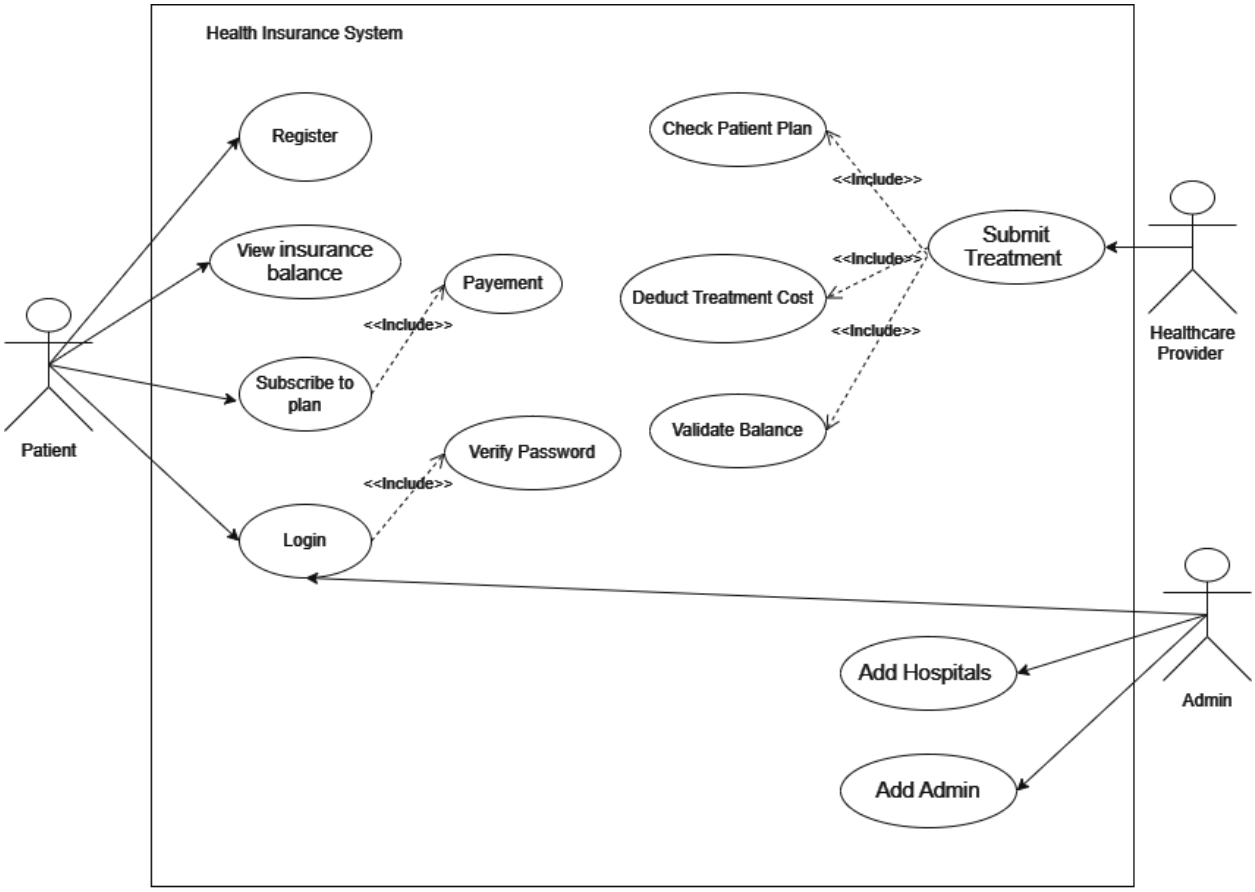
The system interface should be simple and easy to use for patients, Healthcare provider , and administrators.

4.5 Non-Functional Requirement 5: Compatibility

The system should run on standard operating systems and support common development technologies.

5. Sequence Diagram:

Patient Plan Subscription:



6. Use Case Diagram: